

Nestor Guillen

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www.ndguillen.com

Appoinments

Courant Institute, NYU

Visiting Professor of Mathematics

2026 - present

Texas State University

Associate Professor

2022 - 2026

Assistant Professor

2019 - 2022

University of Massachusetts at Amherst

Associate Professor (on leave)

2019 - 2020

Assistant Professor

2014 - 2019

University of California at Los Angeles

E.R. Hedrick Assistant Adjunct Professor

2011 - 2014

Mathematical Sciences Research Institute, Berkeley

Postdoctoral Fellow

2011

Education

Ph.D. in Mathematics

University of Texas at Austin. Advisor: Luis Caffarelli.

December 2010

Licenciado en Matemáticas

Universidad Simón Bolívar. Advisor: Lázaro Recht.

July 2006

Fellowships, grants and awards

Frontiers of Science Award (joint with L. Silvestre).

2026

NSF CAREER Grant DMS-2144232. Award: \$498,969.00

2022-present.

NSF Research Grant DMS-1700307. Award: \$135,000.00

2017-2020.

Visiting Researcher at the *Fields Institute* in Toronto, Canada.

Fall 2014.

NSF Research Grant DMS-1201413. Award: \$102,000.00

2012-2016.

Frank Sid Richardson Foundation Regents Fellowship.

Spring 2007.

Editorial

Electronic Journal of Differential Equations

2022-present

Doctoral students

Cameron Poole.

(Current)

René Cabrera (PhD 2022).

UT Austin.

Michaael Boratko (PhD 2018, co-advised with A. Nahmod).

Google Deepmind.

Selected publications

A complete list of preprints and journal articles can be found in my homepage www.ndguillen.com.

1. **The Landau equation does not blow up**, with L. Silvestre. *Acta Mathematica* (2025).
2. **A Convex Optimization Framework for Regularized Geodesic Distances**, with M. Edelstein, J. Solomon, and M. Ben-Chen. *SIGGRAPH Conference* (2023).
3. **Coupling Levy measures and comparison principles for viscosity solutions**, with C. Mou, and A. Swiech. *Transactions of the American Mathematical Society* (2019).
4. **Pointwise estimates and regularity in geometric optics and other Generated Jacobian Equations**, with J. Kitagawa. *Communications on Pure and Applied Mathematics* (2017).
5. **Estimates for radial solutions of the homogeneous Landau equation with Coulomb potential**, with M. Gualdani. *Analysis and PDE* (2016).
6. **Aleksandrov-Bakelman-Pucci Type Estimates For Integro-Differential Equations**, with R. Schwab. *Archive for Rational Mechanics and Analysis* (2012).

Selected conference talks (2016-present)

- *International Congress of Basic Science (ICBS) 2026*. Beijing, August 2026.
- *Kinetic Theory: Novel Statistical, Stochastic, and Analytical Methods*, Simons Laufer Mathematical Sciences Institute. Berkeley, October 2025.
- *Festum Pi School and Conference*. Crete, July 2025.
- *Science Foo Camp 2023*, Google X. Mountain View, July 2023.
- *ICERM workshop: Optimal Transport in Data Science*. May 2023.
- *V Escuela sobre Análisis Funcional y Geometría*. Buenos Aires (hybrid conference), October 2022.
- *Midwest Geometry Conference*. Plenary speaker. Wichita State University. March 2022.
- *Congreso Latinoamericano de Matemáticos*. (Análisis Funcional y Geometría). September 2021.
- *Gerrymandering workshop*. Duke University, October 2018.
- *Voting Rights Data Institute*. Tufts-MIT, June 2018.
- *BIRS Workshop: Advanced Developments for Surface and Interface Dynamics - Analysis and Computation*. Banff International Research Station. June 2018.
- *Geometry of Redistricting Workshop*. Session on Optimal Transport. August 2017.
- *Mathematical Congress of the Americas*. Session on Nonlocal variational problems. July 2017.
- *SIAM Conference on Nonlinear Waves and Coherent Structures*. Session on Nonlocal dynamics in mechanics, transport, and electromagnetics. Philadelphia. August 2016.
- *3rd Conference on Nonlocal Operators and Partial Differential Equations*. Stefan Banach International Mathematical Center, Bedlewo. June 2016.
- *Conference on Recent Trends on Elliptic Nonlocal Equations*. Fields Institute, Toronto. June 2016

Selected seminar talks (2016-present)

- Computational and Applied Mathematics Colloquium. Pennsylvania State University, February 2025.
- Convexity Seminar. Courant Institute, February 2025.
- Nonlinear Analysis seminar. Rutgers University, February 2025.
- Analysis Seminar. University of Pennsylvania, February 2025.
- Analysis Seminar. Columbia University, October 2024.
- Harmonic Analysis and PDE seminar. CUNY Graduate Center, September 2024.
- Colloquium. University of Texas at San Antonio, March 2024.
- Analysis Seminar. Courant Institute, March 2024.
- Colloquium. University of Arizona, February 2024.
- Partial Differential Equations Seminar. Arizona State University, February 2023.
- Analysis Seminar. UT Austin, January 2023.
- Applied Mathematics Seminar. Brown University. February 2019.
- Colloquium. Texas State University. January 2019.
- Metric Geometry Group Seminar. Tufts University, May 2017.
- Analysis seminar. Massachusetts Institute of Technology, March 2017.
- Analysis and PDE Seminar. Johns Hopkins University. November 2016, Baltimore.
- Analysis Seminar. University of Chicago. October 2016.

Minicourses taught

- 1st San Marcos Winter School, Texas State University, January 2024.
- MIT Summer Geometry Initiative. Faculty mentor, August 2022.
- MIT Summer Geometry Initiative. Faculty mentor, August 2025.
- Minicourse on Stochastic Homogenization. 2016 Gene Golub Summer School, Philadelphia, July 2016.

Selected scientific meetings organized

- [San Marcos Winter School](#), mini school held annually at Texas State since January 2024.
- [Recent Breakthroughs and New Perspectives on the Interplay between Fluid Mechanics and Kinetic Theory](#). Co-organized with W. Golding, D. Albritton, and M. Gualdani, to take place August 2026.
- [AIM workshop on kinetic equations](#). Co-organized with Maria Gualdani, Russell Schwab, and Maja Taskovic. April 2025.
- [Texas State and UT Austin Summer School and Workshop on Algorithms and PDE](#). Co-organized with Maria Gualdani, May 2024.
- [Recent Progress in Kinetic and Integro-Differential Equations](#), November 2022, Banff International Research Station. Co-organized with Maria Gualdani, Russell Schwab, and Maja Taskovic.

- [Workshop on Free Boundary Problems](#), May 2019, Columbia University. Co-organized with Daniela De Silva, Ovidiu Savin, and Hui Yu.
- [Workshop on Nonlinear PDEs](#), December 2018, Columbia University. Co-organized with Daniela De Silva, Ovidiu Savin, and Hui Yu.
- [Generated Jacobian Equations: from Geometric Optics to Economics](#), April 2017, Banff International Research Station. Co-organized with Jun Kitagawa and Robert McCann.
- [Optimal Transport School, Lake Arrowhead \(October 2013\)](#). Co-organized with Dima Shlyakhtenko and Christoph Thiele.
- SIAM Conference on Analysis of PDE 2011. *Non-local equations: perspectives from Probability and PDEs*. Co-organized with Russell Schwab.
- SIAM Conference on Analysis of PDE 2009. *Topics in fractional and geometric PDE*. Co-organized with Luis Caffarelli.

Selected Service

- (at Texas State) Member of the Selection Committee for the 2025 SIAG/APDE Best Paper Prize.
- (at Texas State) Co-organizer of events for Mathematical Awareness Month (2021, 2023).
- (at Texas State) Head writer, proposal for a new mathematics PhD at Texas State. This proposal was submitted to the Texas Higher Education Coordinating Board and the new doctoral program has been approved with a start date of Fall 2025 This was a multiple year effort involving job market analysis, obtaining letters of support from academia and industry, and coordinating curriculum.
- (at Texas State) New PhD Proposal Committee, Chair.
- (at Texas State) Member of: Hiring Committee, Graduate Program Committee, the COVID 2019 pandemic Student Listening Committee, and Committee on Committees.
- (at UMass Amherst) Hiring Committee member (3 searches total between 2014 and 2019), Colloquium committee co-chair (5 years), Graduate admissions committee (3 years).
- (at UMass Amherst) Contributed to the “Amicus Brief of Mathematicians, Law Professors, and Students” submitted by mathematicians and law professors to the Supreme Court for the Rucho V. Common Cause case. March 2019.
- (at UMass Amherst) Panelist for “Gerrymandering steals elections: learn how it’s done and how to stop it”, at Smith College, February 2018. Panelists included members of congress and the state legislature.

Press releases and press coverage related to scholar activities

- Blaschke, Jayme. [NSF CAREER grant fosters research into autonomous systems, increasing math majors](#). Texas State Newsroom (online). February 2022.
- [Texas State to host online seminar on political gerrymandering](#). San Marcos Corridor News (online). September 2020.
- Chan, Dawn. [A summer school for mathematicians fed up with gerrymandering](#). The New Yorker. August 8th, 2017.